Mechanical Properties (Part 3)

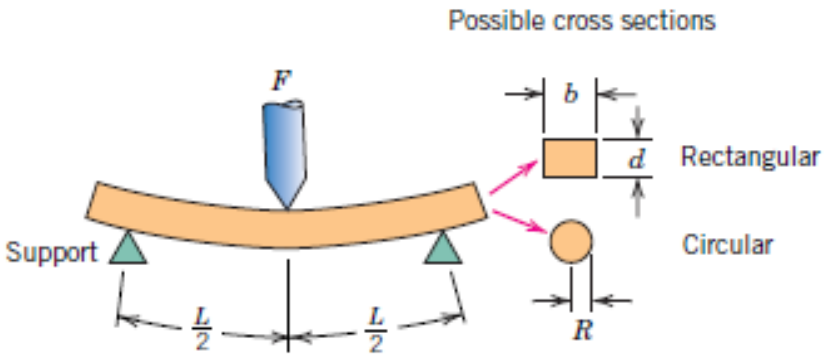


MO-201 Materials Science
Odd Semester 2026-27
(Dr. S. Sinha, Dept. of Met. Engg.)

Reference:
Fundamentals of Materials Science and Engineering
William D. Callister

Testing Brittle Materials

Three point bending test



σ = stress = $\frac{Mc}{I}$

where M = maximum bending moment
c = distance from center of specimen to outer surface
I = moment of inertia of cross section
F = applied load

	M	c	I	σ
Rectangular	$\frac{FL}{4}$	$\frac{d}{2}$	$\frac{bd^3}{12}$	$\frac{3FL}{2bd^2}$
Circular	$\frac{FL}{4}$	R	$\frac{\pi R^4}{4}$	$\frac{FL}{\pi R^3}$

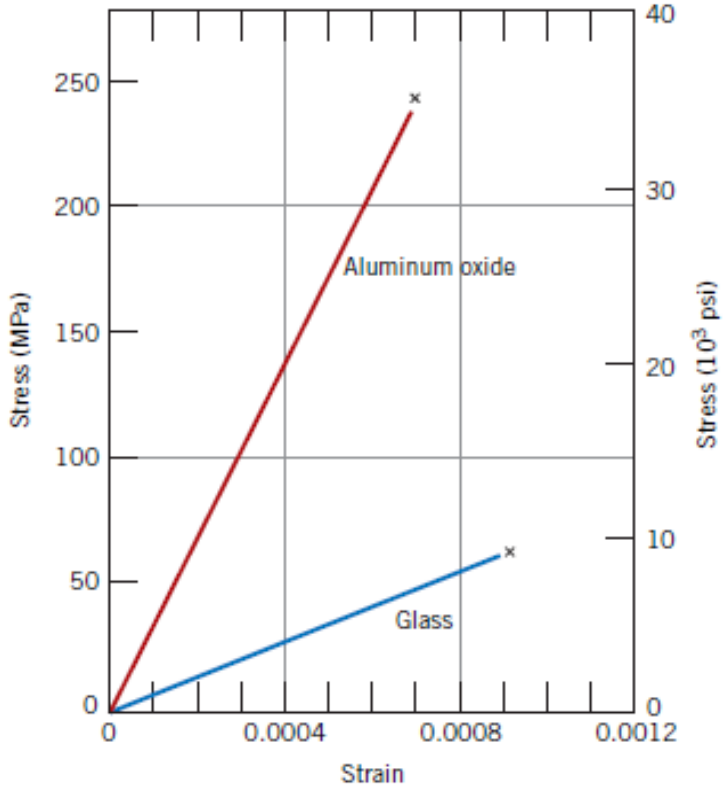
$$\sigma_{fs} = \frac{3F_f L}{2bd^2}$$

$$\sigma_{fs} = \frac{F_f L}{\pi r^3}$$

- Top surface in compression and bottom surface in tension
- Tensile strengths of ceramics are about 1/10th of their compressive strengths, and fracture occurs on the tensile specimen face; hence, the flexure test is a reasonable substitute for the tensile test
- The stress at fracture using this flexure test is known as the flexural strength, modulus of rupture, fracture strength, or bend strength

Testing Brittle Materials

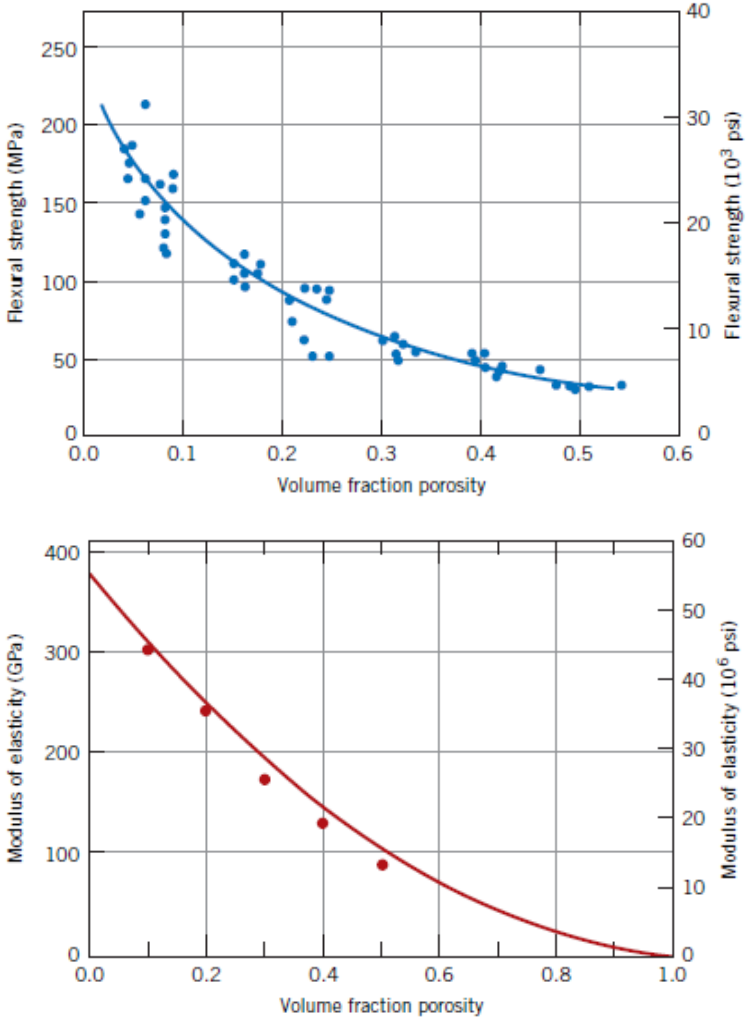
Stress-strain curve for ceramics



$$E = E_0(1 - 1.9P + 0.9P^2)$$

$$\sigma_{fs} = \sigma_0 exp(-nP)$$

Influence of porosity



Hardness Test

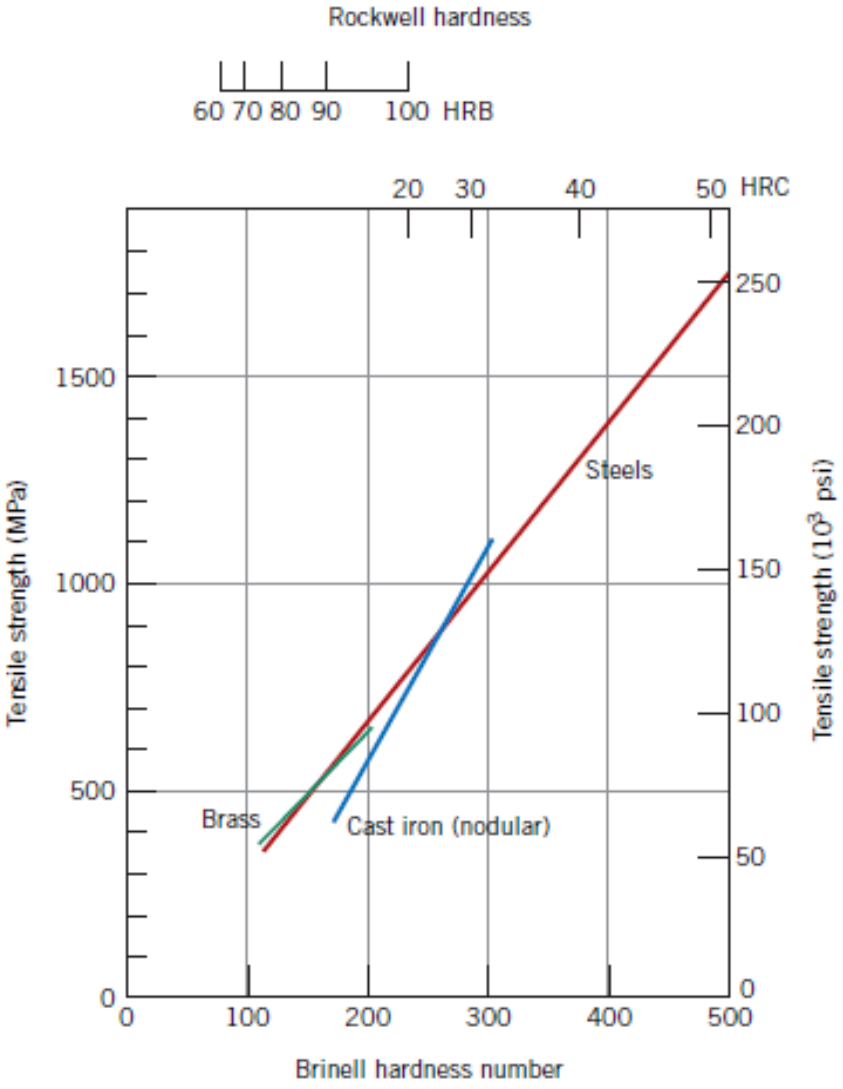
Overview

- Measure of a material's resistance to localized plastic deformation (e.g. a small dent or a scratch)
- Small indenter is forced into the surface of the material under specific load and loading rate
- Depth/Size of the indent is measured to estimate the hardness
- Obviously, softer the material, larger and deeper the indentation
- Remember, higher hardness usually comes with brittleness

Hardness Test

Advantages of Hardness Testing

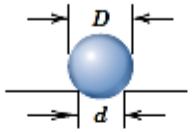
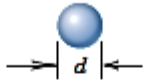
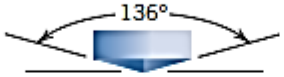
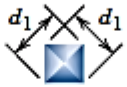
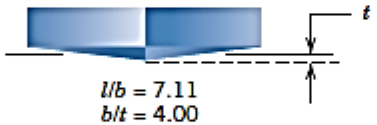
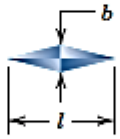
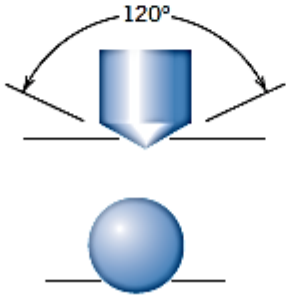
- They are simple and inexpensive—typically, no special specimen need be prepared, and the testing apparatus is relatively inexpensive
- The test is nondestructive—the specimen is neither fractured nor excessively deformed; a small indentation is the only deformation
- Other mechanical properties often may be estimated from hardness data, such as tensile strength; e.g. for steels, TS (MPa) = 3.45 × HB



Hardness Test

Various Hardness Testing Techniques

Table 7.5 Hardness-Testing Techniques

Test	*Indenter*	*Shape of Indentation*		*Load*	*Formula for Hardness Number*[a]
		Side View	*Top View*		
Brinell	10-mm sphere of steel or tungsten carbide			*P*	$HB = \frac{2P}{\pi D\left[D - \sqrt{D^2 - d^2}\right]}$
Vickers microhardness	Diamond pyramid			*P*	$HV = 1.854P/d_1^2$
Knoop microhardness	Diamond pyramid			*P*	$HK = 14.2\ P/l^2$
Rockwell and superficial Rockwell	Diamond cone; $\frac{1}{16}$-, $\frac{1}{8}$-, $\frac{1}{4}$-, $\frac{1}{2}$- in. diameter tungsten carbide spheres			60 kg, 100 kg, 150 kg } Rockwell 15 kg, 30 kg, 45 kg } Superficial Rockwell	

[a]For the hardness formulas given, *P* (the applied load) is in kg and *D*, *d*, d_1, and *l* are all in mm.

Hardness Test

Rockwell Test

- Hardness number is determined by the difference in depth of penetration resulting from the application of an initial minor load followed by a larger major load (the minor load enhances test accuracy)

- For each scale, hardness values up to 130 (but inaccurate < 20 & > 100)

Table 7.6a Rockwell Hardness Scales

Scale Symbol	*Indenter*	*Major Load (kg)*
A	Diamond	60
B	$\frac{1}{16}$-in. ball	100
C	Diamond	150
D	Diamond	100
E	$\frac{1}{8}$-in. ball	100
F	$\frac{1}{16}$-in. ball	60
G	$\frac{1}{16}$-in. ball	150
H	$\frac{1}{8}$-in. ball	60
K	$\frac{1}{8}$-in. ball	150

Table 7.6b Superficial Rockwell Hardness Scales

Scale Symbol	*Indenter*	*Major Load (kg)*
15N	Diamond	15
30N	Diamond	30
45N	Diamond	45
15T	$\frac{1}{16}$-in. ball	15
30T	$\frac{1}{16}$-in. ball	30
45T	$\frac{1}{16}$-in. ball	45
15W	$\frac{1}{8}$-in. ball	15
30W	$\frac{1}{8}$-in. ball	30
45W	$\frac{1}{8}$-in. ball	45

Rockwell Test

Scale symbol	Indenter	Major load (kgf)	Typical applications
A	Diamond (two scales —carbide and steels)	60	Cemented carbides, thin steel, shallow case-hardened steel
B	1.588 mm ball	100	Copper alloys, soft steels, aluminum alloys, malleable iron
C	Diamond	150	Steel, hard cast irons, pearlitic malleable iron, titanium, dee case-hardened steel, other materials harder than HRB 100
D	Diamond	100	Thin steel and medium case-hardened steel and pearlitic malleable iron
E	3.175 mm ball	100	Cast iron, aluminum and magnesium alloys, bearing metal
F	1.588 mm ball	60	Annealed copper alloy, thin soft sheet metals
G	1.588 mm ball	150	Phosphor bronze, beryllium copper, malleable irons. Upper limit HRG 92 to avoid possible flattening of ball
H	3.175 mm ball	60	Aluminum, zinc, lead
K	3.175 mm ball	150	Bearing metals and other very soft or thin materials. Use smallest ball and heaviest load that do not produce anvil effect.

If a diamond spheroconical indenter is used, then

Rockwell Hardness Number = 100 - (H/0.002)

If a ball indenter is used, then

Rockwell Hardness Number = 130 - (H/0.002)

Where 'H' is in mm.

https://theconstructor.org/practical-guide/rockwell-hardness-test/

Hardness Test

Rockwell Test

- Inaccuracies also result if the test specimen is too thin, if an indentation is made too near a specimen edge, or if two indentations are made too close to one another

- Specimen thickness should be at least 10 times the indentation depth, whereas allowance should be made for at least three indentation diameters between the center of one indentation and the specimen edge, or to the center of a second indentation

- Testing of specimens stacked one on top of another is not recommended

- Accuracy is also dependent on the indentation being made into a smooth, flat surface

Hardness Test

Brinell Test

- Standard loads range between 500 and 3000 kg in 500-kg increments; harder materials require greater applied loads
- Load is maintained constant for a specified time (between 10 and 30 s)
- Brinell hardness number, HB, is a function of both the magnitude of the load and the diameter of the resulting indentation (measured with microscope)
- There is only one Brinell scale
- Maximum specimen thickness and indentation position (relative to specimen edges) as well as minimum indentation spacing requirements are the same as for Rockwell tests; also, smooth, flat surface is necessary

Hardness Test

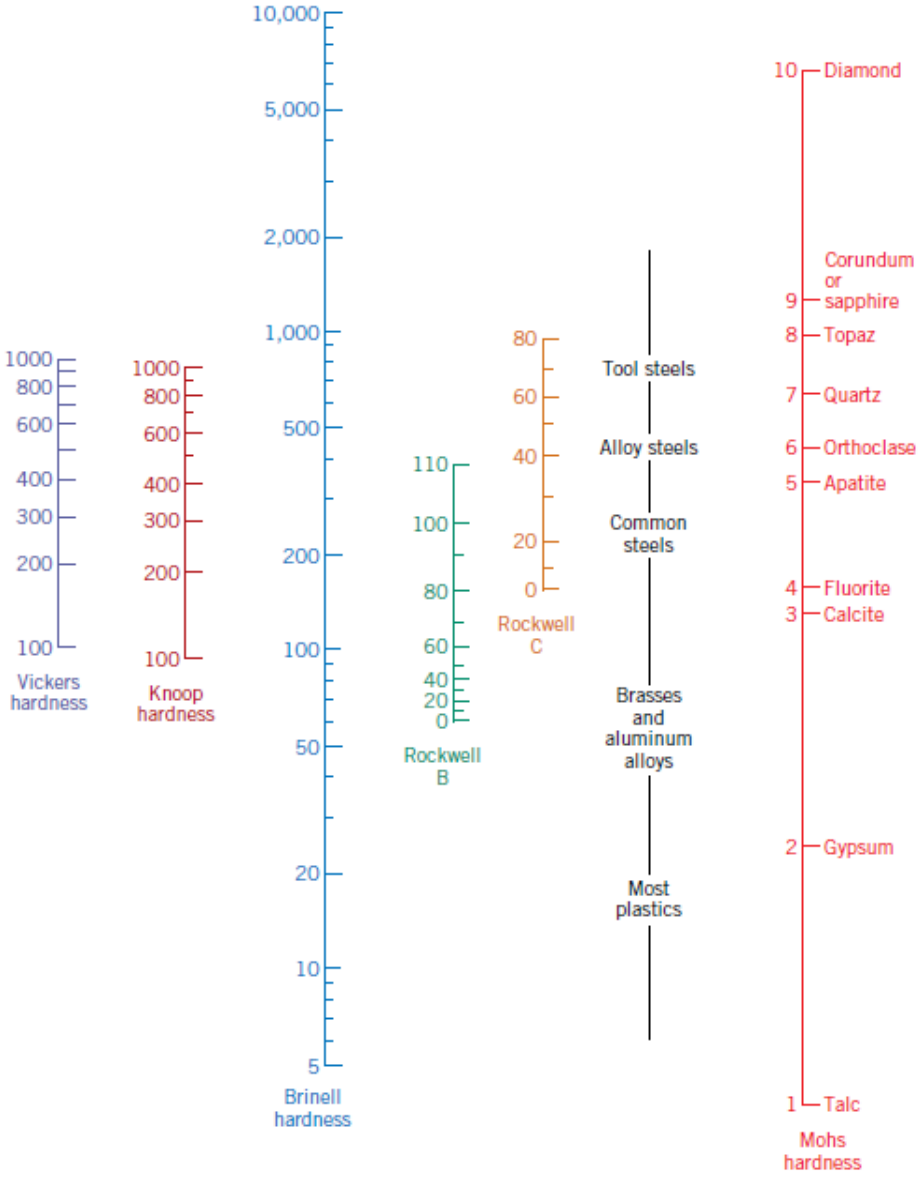
Knoop & Vickers Test

- Very small diamond indenter having pyramidal geometry is forced into the surface of the specimen
- Applied loads are much smaller than for the Rockwell and Brinell tests, ranging between 1 and 1000 g
- The resulting impression is observed under a microscope and measured; this measurement is then converted into a hardness number
- Careful specimen surface preparation (grinding and polishing) may be necessary to ensure a well-defined indentation that may be measured accurately
- The Knoop and Vickers techniques are referred to as microindentation-testing methods on the basis of indenter size

Hardness Test

Hardness conversion

- Because hardness is not a well-defined material property, and because of the experimental dissimilarities among the various techniques, a comprehensive conversion scheme has not been devised
- Hardness conversion data have been determined experimentally and found to be dependent on material type and characteristics
- The most reliable conversion data exist for steels



Hardness Test

Hardness Testing of Ceramics

- Spherical indenters (as with Rockwell and Brinell tests) are normally not used for ceramic materials because they produce severe cracking

- Vickers test is widely used for measuring hardness of ceramics; however, for very brittle ceramic materials, the Knoop test is often preferred

- For both techniques, hardness decreases with increasing load (or indentation size) but ultimately reaches a constant plateau that is independent of load; the value of hardness at this plateau varies from ceramic to ceramic

- An ideal hardness test would use a sufficiently large load that lies near this plateau yet be of magnitude that does not introduce excessive cracking

Hardness Test

Tear Strength and Hardness of Polymers

- The ability to resist tearing is an important property of some plastics, especially those used for thin films in packaging

- Tear strength is the energy required to tear apart a cut specimen of a standard geometry; magnitudes of tensile & tear strengths are related

- Polymers are softer than metals and ceramics; Rockwell tests are frequently used for polymers; other indentation techniques employed are the Durometer and Barcol tests